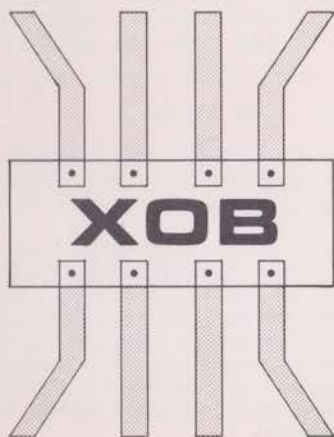


Econet Software



These notes are a supplement to our 'Product Guide' which provides a quick reference to price and compatability and a brief description of each item.

The items listed in this book are arranged in alphabetical order. After first finding an item of interest in our 'Product Guide' you may look up further details in here.

Accelerator

Author : W.Tunstall-Pedoe

This package is designed to speed up file access on Econet systems with BBC Model B computers. It is written by the author of the acclaimed 'Network Operator' software.

This package allows ordinary BBC model B computers the faster Econet file access found with the ANFS supplied in the Master series. By using a buffer in the machine 100 bytes are grabbed at once instead of the usual 1. This technique gives speeds similar to those of the Master series and can be even faster.

This software optimises speed whilst minimising memory usage. A double benefit is gained when several machines are using accelerator since the total load on the network is reduced for all stations.

Accelerator is superior to some utilities which try to achieve the same effect since it allows true random access as well as serial access. INPUT, PRINT, PTR, BPUT and BGET are all supported.

Alternative versions of the code are supplied to allow you to optimise the software for your network. Demonstration software shows how to use the code which is called from the network library.

Anti-Virus

Author : Ian Morrow

Would be hackers tend to protect the machine they are using so that the system manager cannot observe what is happening.

When virus infections are present in machines they tend to take over operating system vectors.

This program checks all BBCB, B+ and Master series machines for tell-tale signs of virus infection based on vector re-direction and reports on effected machines and those which are protected.

Protected machines may cause the system manager to be suspicious and he or she may then chose to examine them in person. Machines suspected of viral attack may be cured remotely and automatically by the software.

This software is designed to deal with the two most likely forms of mischief but it cannot deal with viruses which have transient effects based on disc-based infection.

ASK

Author : W.Tunstall-Pedoe

On issuing the command *ASK the system produces a saying at random from a bank of hundreds supplied. These sayings may be witty, irudite, philosophical or fanciful.

The ASK system provides an element of spontaneous originality which gives a new slant on the capabilities of the network. Users introduced to the ASK system will use it again and again, intrigued by the novelty of the experience. Over two hundred and fifty sayings are included in the starter pack.

Having acted as recipient to the sayings of others the users are then inspired to add their own sayings. Utilities are provided which allow the users to add their own work to a temporary file which may then be checked by a member of staff who may choose to allow each new item to be added to the main file or reject it as unworthy.

ASK may be used for fun, or it may be used as a tool for encouraging original epithets, a starting point for longer works. It helps to give Econet some of the flavour of the facilities found on large main-frame installations

Auto-Backup

Author : Michael Harrison

Another masterpiece from the author of our printer server. This utility will backup complete directory structures automatically between file-servers, or between discs on the same file-server.

More than that it can be used to copy all files or just those which have been altered or added since the last backup. Technically this is known as Incremental Backup.

A utility is included which allows files on the backup disk to be deleted if they do not exist on the source disk.

The effect of the complete package is that a backup disk or file-server can be kept as near exact parallel of the main disk or server with a minimum of effort and network load.

Once you have experienced this system your only regret will be that you did not have it sooner.

Bio-Statistics - The XOB Statistics Package

Author : G.H.Ballinger

The XOB Statistics package incorporates the following tests :-

- (1) Descriptive statistics :-
Mode, Median, Mean, Semi-interquartile range,
Deviance, Standard deviation
- (2) Correlation Tests :-
Kendal's tau, Correlation Coefficient Test
- (3) Weighted Kappa Test
- (4) T test : Independent Samples, Correlated Samples
- (5) Non-parametric tests :-
Mann-Whitney 'U' test, Wilcoxon Test

An arbitrary maximum limit of 1000 samples per file is set but this may be modified if we are notified in advance of supply.

Tests are accessed by means of a menu system which also allows access to a data editor for the entry of new data. Technical information is provided in the documentation to allow users to tailor the output from their own software to the file requirements of the package.

Battleships

Author : David Bainbridge

Using high resolution screens on two Master series computers two combatants play the traditional battleship game, interactively in real time. Moves made at one station are transmitted to the other station and recorded.

An excellent introduction to inter-machine communication and interactive computing.

Multiple pairs of players can use the game simultaneously.

BeebRoms

Author : Geoffrey Ballinger

This gives Model B machines the following commands found on the Master Series machines :-

*ROMS, *INSERT, *UNPLUG

- *ROMS : Shows available ROM software
 - *UNPLUG : Disable chosen language or service ROM
 - *INSERT : Enable chosen language or service ROM
-

There are many networks of the first generation BBC B which do not have several of the utilities now incorporated into the Master Series. On mixed networks it can be frustrating to find that facilities available on later machines are not available on older machines. This package helps redress the balance. Eight commands for the network library.

DUMP : Like the disk utility this dumps a file.

TYPE : This types a file to screen or printer.

CTYPE : This behaves as 'TYPE' but filters control characters which often prevent files being typed.

BUILD : Like the disk utility this allows text files to be built directly from the keyboard.

FSPOL : A faster version of type. This gives the older machines a speed similar to that on the Master Series.

FEP : The Econet system is subject to clandestine viewing of data by using *VIEW or home-grown versions of that program. The traditional *PROT does not give adequate protection on networks with NFS 3.34 or earlier. That allowed privileged stations to override PROT. FEP is a utility which should only be available to the system user and completely protects that user from viewing and interference from others connected to the network.

FEEXEC : Like the EXEC utility but much much faster over the network. Speeds similar to the Master Series.

FTYPE : Like TYPE .. but much much faster over the network. Speeds similar to the Master Series.

CAVE

Author : G.J.Leach

Cave is a true multi-user dungeon adventure. The first to be sold for Acorn's Econet system. It allows users to experience on a micro-computer network the new generation of interactive games previously confined to main-frame machines.

Each Econet station accomodates one of the players. Each player has his or her own identity and password.

On entering the cave each player passes through a number of 'rooms' where he or she may meet computer controlled monsters or other cave users and may accumulate objects with which to enhance prowess.

The doings of any user effect all others. Battles may be fought in real time between users, or users may send messages and provide assistance. From beginnings as a humble caver each user, with skill, may progress by steps to Wizard. If a trip to the cave does not result in the death of the caver then the score of the caver is enhanced and preserved according to the results of any battles fought.

A wide range of commands is available to permit a variety of interactions with cave objects, cave monsters and other cavers.

Cave may be used simply for fun; its many users across the world will testify to the fun of the Cave experience. But Cave is more than a game. It follows the current trend to fitting educational experiences in entertaining packages.

Because it allows immediate interaction between many users and their computers and each other it may be used to demonstrate real-time, on-line, interactive computing. Because it leads to

exciting interactions between users in both co-operative and conflict situations it may be used as a stimulus for social discussions amongst students of all ages. Because it is exciting and imaginative it may stimulate the students that other programs cannot reach.

Response speed is generally better than would be experienced over telephone lines to a main-frame and is particularly fast on level III. Cave may be used on all BBCB and Master series machines.

CAVE-PLUS

Author : G.J.Leach, Enhanced by S.A.Hewick

Building on the success of the original CAVE this upgrade has twice as many rooms, more cave monsters, more cave objects and more complex use of cave objects.

In addition several manager utilities are supplied to monitor the status of the game and maintain the directory of Cave users.

Users of the original Cave have commented that it has to be played to be believed. On trying this latest version they comment that Cave-plus is as far ahead of Cave as Cave is ahead of single user games.

In order to have room for the enhancements Cave-plus requires machines to have Page set at &1200 or lower. This means that it can be used on BBC B computers, Master 128, and ET machines. Unlike the original it cannot be used on BBCB machines which have a DFS fitted, or the Acorn printer server, or teletext adaptor.

On most networks this would simply mean that only certain stations are excluded from use with Cave.

DAY/TIME

Author : M.Ryan

(Was part of Networker's Friend)

TIME : On level three servers this allows the display of current system time. It may be used from within BASIC programs to provide time stamping in user programs, a facility not available with the Acorn version.

DAY : When used with level III systems this returns the current day and month. Like TIME this may be used from within BASIC programs to provide date stamping.

DIRECTORY MANAGER (DirMan)

Author : W.Tunstall-Pedoe
Good for Network Managers

When re-arranging and tidying directories it is inconvenient to have to constantly change between the various utilities needed for this process. Typically a Network Manager would use *ACCESS, *RENAME, *DELETE (or *WIPE), *INFO, *CAT and *DUMP. The use of *WIPE and *DELETE is critical. At any time an error could result in loss of valuable material.

DIRECTORY MANAGER makes the managing of directories a joy. The entire directory contents are visible on screen without the need for repeated *CAT commands. Files may be accessed, renamed, deleted and undeleted with single-key commands. The file parameters (as shown by *INFO) are in view as the file is considered. It is possible to change and re-change your decisions as to the fate of the file and work backwards and forwards through the directory. Only when you are satisfied that all the changes are correct will they be carried out. In this way there is little chance of making a wrong decision in the heat of the moment which will return

to haunt you.

DIRECTORY MANAGER will be useful for any user with files to manage but it is particularly useful for the Network Manager.

DIRMAN+ Authors W.Tunstall-Pedoe and M.J.Sims

The specification for this is as for Directory Manager but the software has been modified to be compatible with the Archimedes.

DUPE Author : G.H.Ballinger

This utility allows the duplication of files within the directory structure of the network.

e.g. *DUPE \$.hold.!BOOT \$.S1.Fred.!BOOT

This would create a copy of \$.hold.!BOOT in S1.Fred

FASE Author : W. Tunstall-Pedoe

This is the first commercially available encryption system for data stored on Econet systems.

The FASE software caters for the automatic encryption and decryption of data on BBC/Acorn computers using Acorn's Econet system. It allows documents, data files and programs to be stored in a coded form on the network so that they may not be read or used by other users even in the event of a breach of normal security. It provides a new dimension in data security for Econet systems.

Two stages of encryption are supported.

At the first level of encryption data is coded in a preliminary fashion so that it may not be interpreted during transmission over the network, nor during casual reading of files by unauthorized personnel. This encryption is transparent to the user, as is the decryption. Full random file access and serial access are supported and the system can actually speed up file transfer on older systems.

Even greater security is provided by software which allows existing files, encrypted by the first level or in clear form, to be further encrypted using a sophisticated encryption technique. This second level permits data to be secure against more determined attempts to breach security.

FINDTHEM Author : S. Hewick

It is easy to lose files in a complex directory structure. It is also possible for users to make illegal copies of software into their own directories. This utility allows software to be located even if the name is changed or is unknown.

The user specifies a starting root directory and what ever information is known about the file's addresses and length, leaving other information as 'wild'. FINDTHEM then works down through the directory structure listing all objects conforming to the parameters given, with complete path names for each. Options permit this list to appear on the VDU only or on VDU and printer.

FRIENDLY

Author : W.Tunstall-Pedoe
Was part of Networker's Friend

This consists of a set of programs which may be added to the Econet File-Server disk. The effect of the package is to make the system more user-friendly and secure.

(1) Password entry program

This is a user-friendly logging-on program. The user simply presses shift & break. The program presents the user with a customised screen bearing the institutions title.

The user is first asked for his/her user name and then for the password. The password is masked from view on entry. Help is available at each stage.

The program is not only less intimidating than the *I AM sequence but provides a layer of protection since it inhibits 'trying out passwords'. Because it is a 'machine code' program it reduces the chance of users breaking in to the boot directory. Level two file-servers may be crashed by attempts to enter passwords of more than six characters. This program protects against this.

(2) Log-off program

When run this does a *BYE and then displays a sequence of constantly varying screen images. These not only prevent selective 'burning out' of screen phosphors but also periodically display a message showing the institution title and how to operate the network boot sequence.

(3) Menu

A BASIC program which may be placed in any teaching directory to allow programs to be chosen by a single keypress. This is customised to the institution.

(4) PSET

No more complicated setting up for your network printer. Just type *PSET and the network printer is set up complete with customised banner. What is more it remains set-up even if BREAK or CTRL-BREAK are pressed. Versions supplied for both serial and parallel printers.

Each item in this package displays a header customised to your institution and department title. This should be specified when ordering the software. Apart from the friendliness and security improvements realised by using these utilities they give your system a professional gloss.

FSCOPY

Author : M.J.Sims

Archiving is a big hassle on most systems with large file-storage. Many people feel that tape streamers are unreliable or cannot adequately be verified. Some users have decided to use secondary winchester disks as backup, either on a second file-server linked through the net with the primary machine, or to a second disk connected to the primary file-server. This utility copies entire tree structures, or sections of tree-structure flexibly between discs on the same file-server, or between file-servers on the network.

By minimising code size and dedicating the software to this single task the buffer size is maximised to shorten copying time.

See also TreeCopy

GET

Author : W.Tunstall-Pedoe

Isn't it inconvenient to have complicated path names leading to often used software ? It would be better to keep such software somewhere where anyone can get at it without being able to damage it ? Putting programs in the Library would be ideal; but then they are usually difficult to get at. GET changes all that.

e.g. CHAIN "\$.LIBRARY.GRAPHICS" - becomes - *GET GRAPHICS
CHAIN "\$.LIBRARY.Year1.Demo" - can become - *GET Year1.DEMO

You can write your own utilities in BASIC and put them in the LIBRARY too.

GETPASS

Author : S.A.Hewick

SYSTEM MANAGER ONLY
NOT S.J. Systems

When a user forgets his or her password it is inconvenient to have to use *REMUSER and re-instate with *NEWUSER. In doing this the BOOT option is lost together with existing space allocation (Level III). In addition it is sometimes necessary to examine the system from the point of view of a particular user when the user is not available.

This utility is used in the form *GETPASS <user id.>. If the user specified exists and does not have system status that users password is returned. If the user does not exist this fact is indicated. If the user has system status then this is stated but no password is returned.

GIVE & TAKE

Author : M.J.Sims

SYSTEM MANAGER ONLY

These utilities allow programs to be distributed to network machines and set running, or captured from network machines to the local machine.

GIVE passes a program from the local machine to a group of specified machines on the network. All the machines may thus be set up to run the specified program, and the file-server need not even be active. This means that the machines need not be logged on to the network, it also means that software can be distributed even if the file server has failed.

TAKE is capable of taking software from a chosen machine or group of machines sequentially and saving the software to local disk. It may be used for examining the user's program in order to offer advice, or for saving programs in the event of a failure of the network file-server.

GRABBER

Author : G.H.Ballinger

This software allows the automatic capture of teletext pages for use with the CommunTel viewdata system supplied with Level III and FileStore. It may be used directly with the Acorn Teletext server or, if you have the XOB TSERVER package it may be used remotely from another network station.

The software has its own easily mastered control language to permit page grabbing from any channel, automatic overlay of text on pages, and the automatic routing of pages.

Help-on Author W.Tunstall-Pedoe

Master series computers have a 'HELPON' facility which allows files of helpful information to be set up in the library directory. *HELPON gives a list of the help categories available. *HELPON (topic> gives help on that topic. This utility provides BBCB machines with the same facility.

Log-off Author W.Tunstall-Pedoe

Specify a user, or a group user and that user, or all users with that name will be logged-off.

Menu System Author : S.A.Hewick

Users require access to large amounts of software yet many may have little knowledge of even simple of commands.(e.g. The difference between CHAIN(BASIC), RUN(Comal) and *RUN.

Teaching staff with little computing knowledge should NOT feel ignorant when attempting to use networks. Network managers should NOT need to spend ages setting up specially tailored menus.

Netmen is the solution. It used in any user directory where software is to be accessed in the simplest possible manner. It may be stored as the !BOOT file in a given user directory and the option set to RUN, or it may be called by other software.

The user is presented with a menu of FILES WHICH ARE DIRECTLY EXECUTABLE. Confusing text-files, screen images and machine-code procedures are NOT shown. Though many programs may be accessed in this way only 7 are presented at one time to avoid confusing the user. The space-bar is used to call up successive lists of 7 programs so the number of programs available from a single menu is as large could reasonably be needed. Any item of software may be selected by a single-key entry. The menu program automatically selects BASIC or COMAL, or machine code as appropriate and executes the program accordingly. The system adapts as new software is added and the manager incorporates the new software simply by saving it in the appropriate directory. Suspension of options is equally simple.

Netmen is is fast and can be used to disable break and escape during program choice. NetMen makes life simple for the user and simple for the system manager.

NETLOAD Author : W.Tunstall-Pedoe

As networks become larger it becomes important to be able to predict the times when they will be heavily loaded and the times when there is spare capacity. This utility produces a graphical representation of genuine network load without itself effecting that load. The display indicates load versus time permitting the Network Manager to investigate the times when file transfer delays are likely to be noticed, or when access will be particularly rapid.

This software allows a spare network station may be used to run a series of programs automatically. These may allow the provision of special services at particular times, or commence certain security procedures automatically. In effect, when used with other software, it allows the Network Manager to be present when absent.

NETMAIL

Authors : G.H.Ballinger & G.Leach

ELECTRONIC MAIL SYSTEM

This electronic mail system allows some or all users to send mail messages to all other users of the same system or to a single user. The system manager may send messages to all other users, or to a single user. It is possible to configure the system to allow free access to all users, or to restrict access in a variety of ways.

The XOB mail system maximises friendliness and flexibility. Some mail systems require a machine to be dedicated to serving the mail. This system does not. The NETMAIL software is always available as long as the normal file-server is present.

Some systems restrict the number of users who may use the electronic mail. This system allows all normal users to have access to electronic mail.

To reduce the likelihood of 'junk' mail every item is stamped with the date and time of sending and carries the identity of the sender. This identity is checked when the mail is being sent.

Three categories of mail are recognised :-

MAIL is system mail. It is sent to every user but can only be written by system users. When the *GETMAIL command is used the user MUST read this mail before being allowed to exit.

GENERAL MAIL is mail which may be written by any user and read by every user.

PERSONAL MAIL may be written by any user and read only by the user To whom it is addressed.

ALLOWED is a file of users who are allowed to SEND mail. If this file is not present then ALL users will be allowed to send mail otherwise the user must have his or her name present in that file if they wish to send mail. The ALLOWED file may be edited using the program called EDALLOW or with a word-processor if one is available.

Users may be forced to read any mail which has accumulated by building the *GETMAIL command into the BOOT sequence. Alternatively *GETMAIL may be called at any time.

Personal mail may be read, sorted and printed at any time using *SORTMAIL.

Mail may be sent to individuals or to ALL USERS using *SENDMAIL

The System Manager may generate mail in a variety of ways :-

By the usual *SENDMAIL command.

By spooling text from a word-processor.

By *SAVING Mode 7 screens from BASIC. This allows coloured text and even graphics to be sent.

Both public and private mail can extend over many screens. At the end of each screen or message the system waits for a key to be pressed before displaying the next section.

NOTICEBOARD

Author : M.Ryan

Carousel systems used with many viewdata-systems lack flexibility. This one may take pages from any viewdata system compatible with Econet and displays them in any chosen order, you are not limited to 'a' pages, or by location of the pages within your data-base structure. Special pages may be made to appear at pre-arranged times. Please contact before ordering so that we may ensure the package works with your favorite viewdata system.

NETROMS

Author : G.H.Ballinger

The Master series computers mean that ,with appropriate license agreements, it is now possible to keep vast quantities of ROM style software in the network library. Normally there is one serious drawback, the process of loading and installing this software involves the user in several tricky and long-winded commands culminating in having to perform a CTRL-BREAK.

NETROMS makes loading such software as easy as it can be. It allows you to customise your 16k ROM software so that it may be called, installed and set working with a single command.

Do *COMAL - you are in COMAL, *WORDWISE - WORDWISE appears.

If you can port the ROM to the Library you can call it with the NETROMS system. You can even change the name if you want. If you have a ROM image of PASCAL you can arrange to call it with *PASCAL, or *QUICHE.

NETROMS helps you to realise the full potential of networked Masters.

NETWORK OPERATOR II Author : W.Tunstall-Pedoe
HIGH SECURITY PROGRAM
Upgrade from NETWORK OPERATOR

Experience with networks in teaching situations shows that the network supervisor needs immediate and effective control of remote stations both individually and in groups. Utilities supplied with the Econet system provide insufficient help. The individual utilities sold by some publishers are mere toys in comparison with this.

In particular it is essential for the network supervisor to be able to achieve these ends :-

(1) To communicate effectively and with the correct amount of interference when demonstrating points and providing help.

(2) To regulate those bent on mischief with minimum disturbance to other users and with maximum ease for himself (herself).

(3) To perform routine housekeeping without having to visit every station on the network.

Network Operator permits you to do all of these things. At any time you can control not only what is happening at stations which are logged-on, but any station which retains an intact link with the Econet.

Operational Conditions

An Econet system with any level of file-server software.

The Network Operator software may be used with OR WITHOUT the file server operational. Software is supplied on disk and may be loaded into the network from the file server from standard disc system or from tape.

The 'Station Table'

Any command may be addressed to a group of stations by using the 'station table'. Alternative station tables may be set up and stored to be used as required. Individual stations may be included in or discluded from the station table whilst the Net Operator is running and the software can automatically detect which stations are on the network and automatically build its own station table.

Typical Command Sequences

(a) Warning all stations that the network is about to close down (or start up), or of imminent disk changes.

(b) Correcting student forgetfulness

e.g. Logging off all stations or selected stations no longer being used. Blanking VDU displays on unused stations to save monitor phosphors, or avoid distractions.

(c) Gaining control of rogue stations.

e.g. Forcing stations running illegal programs to run legal programs. Halting all stations without damage to resident programs whilst some teaching point is made.

(d) Demonstrating teaching points with minimal disturbance to on-going work.

(e) Sending the same program to a group of stations and ensuring all stations execute that program at the same pace.

Unlike the *REMOTE command which is somewhat brutal, Network Operator allows many levels of control ranging from the underhand through the subtle to the exceedingly brutal.

The Macro facility

The commands used by Network Operator are single-key commands. Extremely complex sequences of commands may be stored on the user-defined keys. Alternative key definitions may be stored and it is thus possible to have a library of frequently used command sequences which may be implemented almost instantaneously.

Network Programming Language

Network Operator may be called from within BASIC programs and used as a network programming language. BASIC programs may be simply written which execute very complex operations on the network. By running a single program different stations may be set to the same or different tasks. The entire network may be re-configured by a single command. Demonstration software supplied shows how.

Space does not permit all commands to be listed but they include :-

B Blank	Totally clears screen of target station.
FB ForceBreak	Causes station to do a soft reset.
FC Force Command	Passes a command to a station exactly as if it had been typed at that station (e.g. *I AM DEMO)
FH Force Hard	Station does hard reset.
FO Force Osbyte	Causes remote 'OSBYTE' call. (*FX)
FR Force Reset	Logs the station off the network and completely clears the stations
FS Force Shift-break	Causes remote auto-boot.
GM Give Memory	Send section of your memory to another machine.

L	Log-off	Causes a soft reset and *BYE the target station.
N	Notify	Sends message to remote station. Better than *Notify since this does not interfere with typing which is on-going.
Q	Quieten	De-activates sound output.
TM	Take Memory	Exact opposite of Give Memory. May be used to salvaging a program in the event of a file-server failure. (Program captured to ,say, a DNFS machine from an NFS machine.)

No other utility for Econet, from any publisher, has the power and flexibility of Network Operator II. No other utility for Econet is as simple and swift in use. All commands are logical, but extensive help facilities are built in for the forgetful. Speed and simplicity is assured through single key entry of commands (the software fills in what you miss out).

More than 30 fundamental commands may be combined in limitless ways and the package may be used across bridges and comes with a licence which allows it to be used on up to five linked networks.

OAKTREE

Author : M.J.Sims

As directory structures become more complex, and user populations large, it becomes increasingly difficult to assess the nature of the complete structure and move readily through the structure to do security checks, look for wasted space or find particular items. Paper printouts of the tree-structure take time to generate, are heavy on paper-consumption and ultimately inflexible.

OAKTREE allows you to see the structure of your network directories on screen (or in clearly understood form on Epson compatible printer) and to move through that structure at will examining directories and files.

PassMan

Authors : M.Ryan & W.Tunstall-Pedoe

NOT S.J.Systems

SYSTEM MANAGERS ONLY

The password management system allows the manager to keep track of all the users registered on the system.

On level II systems the following details may be displayed :

All users or only system users, with or without passwords.

Boot option setting for all users.

The list shown may be in original order or alphabetical.

It is possible to search for a user-name.

The data may be printed to screen, printer, or filed.

Filed data may be used to facilitate changing the user data in the password file using software supplied, or written by the user.

On level III systems all of the above are possible and the free-space allocation of users is displayed. Groups of users may be searched for.

The status of the entire user population may be viewed at a glance and regular checks are possible for users who have illegal options, or who may have illegal system status. Software is included in the pack to illustrate a range of applications of the data gathered (e.g. setting up options for entire groups of users).

PASSWORD CHANGE (PWC) Author : W.Tunstall-Pedoe
Was in Networker's Friend

PWC is a machine code utility which allows users to change their password without other users being able to see it. Anyone may use it at any time. The screen display is clear and friendly and the new password, though masked, is entered twice so that it is verified before it is changed. Because users know that their password cannot be seen as they change it they tend to change their passwords more often giving a higher degree of security, particularly important for System Users.

SEARCHER Author : S.A.Hewick

SEARCHER is a major advance in information retrieval on Econet systems. It allows text files to be set up on a variety of topics and searched on a key-word basis for information on a particular topic. The files to be searched may be an individual user's own files or those common to all users.

Files of book references, telephone numbers, vocabulary and the like and use searcher to find a given telephone number, books on a given topic, or the meaning of a particular word.

On a system basis the same options are possible, or the files may contain information about the system options, locations of resources or staff or schedules of work.

Like most good packages the scope of SEARCHER is limited only by the imagination of the user and the memory capacity of the system supporting it. All the files may be generated by a standard word-processor with spooling facility, or even from the spooled output of BASIC programs.

SELF-CUSTOM Author : G.H.Ballinger

This allows the user to make up a file which will modify the normal Acorn message displayed when BREAK is pressed. The resulting utility may be called from the library in the BOOT sequence to provide a degree of system customisation.

SHUTDOWN Author : E. MacFarlane-Smith

Users may be logged off remotely (unless protected) in a very flexible manner. The user may be specified explicitly or groups of users may be logged off using wild cards, or a station, or range of stations may be logged off.

SYSTEM Author : W.Tunstall-Pedoe

Performs an automatic check of the password-file and deletes system users who do not appear in a list of authorised system users. By keeping the authorised list on disk, kept locked away, and performing the easy automatic check at regular intervals system security is greatly increased. Hourly checks are quite feasible.

SWITCH (*SW) Author : S.A.Hewick

It is frequently necessary to move rapidly backwards and forwards between your root directory and another. *SW allows this.

e.g. You own \$.FRED (your User Root Directory)

You are in \$.FRED.DEMO.PICS

*SW

Now you are in \$.FRED

*SW

Now you are in \$.FRED.DEMO.PICS

*SW

Now you are in \$.FRED etc.

THE PRINTER SERVER

Author : M.Harrison

From an author already acclaimed for his advanced graphics demonstrations for Archimedes we present The Printer Server for Econet.

Traditionally a printer server only allows one user at a time to have control of the printer so other users have to wait until it becomes free before starting to print. The XOB printer server overcomes this limitation by allowing up to 16 users to send printouts simultaneously. These print jobs are then queued up, and printed one after the other. This process is often called 'spooling' and it happens without the user being aware of any delay. As far as any individual user is concerned that user has exclusive use of the printer.

The XOB server uses all the memory in the machine it is running on to store queued printouts, (or 'jobs') and if the memory fills up, jobs are temporarily stored on the fileserver. For this reason the printer server is used in a way analogous to the way the file-server is used to send data to and from the disc. Serial and parallel printers are supported as are the latest network features allowing printers to be called by name.

The XOB server can thus replace up to 16 conventional servers or, on average, reduce the users perceived delay in printing to anything up to a sixteenth of that experienced with a single user printer server. This means that the server can be started rapidly and automatically at those times when heavy printing requirements are needed and disabled and used as a normal network station at other times. Unlike the conventional printer server there are no memory penalties when the server is used as a normal station. It also removes the need for adding expensive printer buffers to your system.

The server may be flexibly configured according to the instructions in the comprehensive manual supplied or used with the default settings incorporated. Automatic date stamping and special headings and footings are available and these may be readily changed with the simple menu-driven utilities supplied.

The server can be started and controlled at the server station or started and controlled at another station by the system manager.

Please examine the alternatives, we think you will admit that the XOB printer server is quite simply the best.

TOTAL

Author : S.A.Hewick

The complexity of the Econet system is such that it is important to be able to keep track of the storage space used by each user to ensure that no user is making unreasonable demands on resources.

On level III systems it is possible to read the amount of space left to each user (using RDFREE) but this gives no indication of how much space has already been taken. The situation is complicated further by the way space is debited and credited. It is possible for user A to gain extra space when another user (B) saves data into A's space. Similarly A may lose some of the space allocated by saving into B's directory.

This utility helps System Managers to cope with these problems.

On level II systems this utility allows a rapid assessment of the amount of space taken up by a given directory. If a root directory is specified the space taken by all files and sub-directories is calculated and displayed. This allows rapid assessment of which users are exceeding the desirable space allocation.

On level III systems TOTAL reveals its true power :-

SYSTEM MODE : If the user calling TOTAL has system status the utility displays the space taken by the specified directory. If the directory specified is the root directory of a user then it also displays the free-space left to that user and the total space used and allocated to that user. In this mode it gives a rapid assessment of the storage actually being used and capable of being used.

PUBLIC MODE : An ordinary user calling TOTAL is able to see the space taken up by files and directories from a given root directory downwards.

TO/RETURN

Author : S.A.Hewick

It is often cumbersome to move between directories using the *DIR command. The utilities TO and RETURN allow you to move between a pair of directories without having to specify complicated path names, and without losing your ownership.

The 'old' way

The new way

From : \$.FRED.WORK (and you own FRED)

*DIR (to retain ownership)

*DIR DEMO.PICS

*TO \$.DEMO.PICS

*DIR

*DIR FRED.work

Now you are in \$.FRED.DEMO.PICS

*RET

Now you are in \$.FRED.WORK again
(A SAVING OF 21 KEY PRESSES)

TREECOPY

Authors : M.J.Sims & R.G.Dhir

Copying files is only part of the solution to archiving. Preserving the structure of the original and allowing easy access to archived material is also important. There is a need for a utility which allows network material to be transferred to and from other filing systems preserving structure as far as possible.

This utility allows complete tree structures to be moved between any combination of Econet, ADFS and Disc filing systems. It is particularly useful with Master series machines in that it permits sections of the Network file structure to be moved to ADFS for use on stand-alone systems and, after editing, moved back again if desired. Within the normal limitations of the DISC or ADFS systems the Network Manager or other users may thus edit the actual structure on a stand alone system when a network is not available.

Packages bought in on DISC or ADFS may be transferred to the network with their structure preserved.

See also FSCOPY

TREE-CLEAN

Authors : G.J.Leach & M.J.Sims

Handle Carefully

The larger the file-storage available to the network the greater the tendency for outdated material to accumulate. This is particularly likely with large numbers of users and as users come and go year by year.

File storage is expensive. With large file-stores the cost of wasted space represents an investment unrealised. Efficient usage of space makes it possible to postpone investment in larger file-storage, and makes for efficient usage of the network.

TREE-CLEAN allows automatic deletion of outdated files. It allows a variety of options so that the system manager can set default states. These allow checks to be made before deletion and files to be migrated from network to floppy as an added security against accidental erasure.

TREE-CLEAN may be set to print to paper or screen. It may be set to show outdated files without deleting them. In this way it is possible for the network manager to check on the number of outdated files on the system, the users responsible for wasted space, and to estimate the likely need for clearing the outdated files.

If this technique is to be used then the user must be provided with a means to make sure that files do not get deleted accidentally. Loading and re-saving each file is effective, but loads the network extensively and is cumbersome, particularly with machine-code and data files. *UPDATE efficiently assigns the current system date to any file specified. It may be used on any type of file

TREE-CLEAN/UPDATE is a substantial utility which allows the Network Manager to make best use of the resources available and maintain these in a professional manner.

TSERV**Author : W.Tunstall-Pedoe**

From the author who wrote the highly respected 'Network Operator' at last a genuine teletext server. If you have one BBC computer with a teletext adaptor every machine on your network can have teletext access at the same time.

The TSERV software allows one machine to serve the teletext adaptor in a similar way to the way the file server serves the disc units. The software is written in a manner which allows highly efficient transmission to all machines. By use of an intelligent queueing system the teletext queue is kept to a minimum with the result that the user will not notice any appreciable delay over that normally expected with teletext.

The function keys are used on each terminal to make selection of channels and pages as easy as possible. All four channels are accessible with hold-page, update, reveal, conceal, save frame, load frame and page and channel select.

If you have a 20 terminal Econet you can have up to 19 stations with teletext. Once the TSERVER software is loaded the machines need not be logged on to the file-server. Public stations can have access to the teletext service without leaving them open as users.

This software works with all standard adaptors (Acorn, Morley and GIS) and works across bridges. Two versions are available to deal with non Archimedes networks or those with mixed BBC/Master/Archimedes networks.

VDE (Viewdata-Editor) Author : Dominic Seymour

This software provides Archimedes machines with a powerful Mode 7 screen editor allowing data entry by keyboard and mouse.

Using this package Mode 7 graphics can be created easily and intuitively by movements of the mouse. Even pixel by pixel editing is catered for. Text is entered by positioning the cursor with the mouse and then using the keyboard.

Screen areas may be copied or moved using simple movements of the mouse.

The resulting screens may be saved as Mode 7 screen images compatible with BBC/Master and Archimedes machines, or in CommunITel viewdata format for use with viewdata systems using that standard, including our own system.

To facilitate use on viewdata systems a routing editor is included.

VIEW-ALL**Author : S.A.Hewick**

It is often useful to be able to view each station on the network in turn automatically. This permits the system manager to keep track of potential trouble-makers who may be working on machines out of direct line of sight and even in other rooms or buildings.

Unless this is done with care it is easy to raise the network load to a level where the actions of users are impaired. This utility uses an efficient method of sensing the machines connected to the network and showing each one in turn. If machines are switched on or off the software takes account of this automatically.

Screen mode is automatically sensed and compensated for even on Master series machines.

VIEWDATA

Author : M.Ryan

This package arises from the desire by system managers to have a single system which will provide viewdata services to mixed networks of BBCB, Master and Archimedes machines.

The protocols used are compatible with the popular CommuIntel system and the XOB viewdata system may be used in parallel with that system to provide Archimedes machines with viewdata access, or may be used on its own to provide all machines with those facilities.

VIEWER

Author : S.A.Hewick

The *VIEW utility supplied with Econet systems cannot be used for repeatedly viewing stations unless repeatedly called from the file-server. It is also crude in the way it deals with machines which are not in the same screen mode as the viewing machine.

VIEWER allows the System Manager to repeatedly view other stations with a specified time delay of between 0 and 30 seconds, or on a single instance. Once installed it remains active, or capable of being active until the user disables it. It may be called from the file server, or from disk. It does not require the file server to be operational. It may be used on a cyclic basis to view all stations on the network.

VIEWER automatically senses the screen mode of the station being viewed and switches to the appropriate mode as required even for Master machines.

Wipe

Author : W.Tunstall-Pedoe

Like the disk utility this wipes files from the chosen directory. Each file is presented in turn and is deleted or not depending on the choice made.

The utility is particularly useful on model B machines which do not have DFS fitted along with their NFS.

Master series machines with ANFS do have a WIPE utility but this does not work on ADFS. For such machines this XOB utility provides WIPE for use on ADFS if kept on the ADFS discs.

> Author : W.Tunstall-Pedoe

The most annoying typing mistake. How many times have you typed *> when you intended to type *. (i.e. the short form for *CAT)

This mistake need trouble you no more.

> is kept in the library of the network (or on DFS and ADFS discs). The accidental mistype may still occur but with this utility present you still get the catalogue you intended to request.

Updates, Improvements and New Software

We want you to be satisfied with your purchases from XOB. Whilst it is impossible to please everyone all the time we do take note of comments made by customers and welcome such comments.

If you cannot find a utility you want then let us know ... we may find it worthwhile to develop the utility.

If there is some aspect of one of our products which you dislike, or there is a feature which particularly appeals, then please let us know. Our software will only be what you want it to be if you tell us what you want it to be.

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